

Nathan Tsao

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AI Engineer with experience training models, deploying AI agents, and controlling robots.

EDUCATION

University of Texas at Austin

Aug 2023 – May 2025

Masters of Science: Engineering (Machine Learning & AI)

GPA: 3.74

Thesis: Neural Port-Hamiltonian Differential Algebraic Equations

University of Illinois Urbana-Champaign

Aug 2019 – May 2022

Bachelors of Science: Mechanical Engineering

GPA: 3.87

SKILLS

Machine Learning: PyTorch | Linux | JAX | LangGraph | Hugging Face | FastAPI | Pandas | Ray (RLlib)

ML Systems: FastAPI | AWS | Docker | Distributed Training (DDP)

Programming: Python | TypeScript | SQL | C++

Relevant Coursework: Reinforcement Learning | Statistical Machine Learning | Theoretical Statistics

Languages: English | Mandarin

WORK EXPERIENCE

NASA Ames Research Center

June 2025 – Aug 2025

Autonomous Aircraft Operations Research Intern

Mountain View, CA

- Built a multi-agent reinforcement learning framework (Ray RLlib) for autonomous aircraft routing, reducing average travel time across agents by 30% in simulation.

Berkeley Lights

June 2022 – Sep 2022

Hardware Engineering Intern

Berkeley, CA

- Built Python-based data acquisition and calibration pipelines for OptoSelect chip.

SELECTED PROJECTS

Murcanti

May 2025

Personal Project

murcanti.com

- Designed and deployed an LLM-powered agentic AI chatbot (AWS Bedrock, FastAPI) to intelligently recommend products (RAG) and facilitate checkout (Stripe).
- Deployed custom RESTful API services (TypeScript, FastAPI, Render) for product search and transactions.
- Developed full-stack e-commerce website and iOS application using Next.js, Flutter, Supabase, SQL.

MetaAgent: Text Instructions to Agentic Workflows

Nov 2025

Personal Project

[GitHub](https://github.com)

- Developed a LangGraph agentic workflow that automatically processes text documents (e.g. standard operating procedures) into executable AI workflow Python code in LangGraph.
- Demonstrated that the MetaAgent outperforms the baseline ReAct agent on task succession by 30% on a benchmark content flagging workflow.

YourInvited: Wedding Website Builder

Nov 2025

Personal Project

yourinvited.dev

- Simple website builder built with Next.js, TypeScript backend, and Supabase.

Causal Diffusion Guidance

May 2025

Statistical Machine Learning Final Project

- Developed a diffusion-based training framework for generating causally consistent counterfactuals images.
- Trained diffusion models over multiple GPUs using PyTorch Distributed Data Parallel (DDP).

RESEARCH EXPERIENCE

Autonomous Systems Group: UT Austin

Dec 2023 – May 2025

Graduate Research Assistant

Austin, TX

- Designed and implemented novel physics-informed neural network architectures in JAX for digital twins, achieving $>10\times$ accuracy improvements over baseline deep learning models.
- Developed a low-power human activity recognition model optimized for batteryless sensors using PyTorch, resulting in 15-50% relative improvement over baselines.

Hybrid Robotics Group: UC Berkeley

May 2022 – Jan 2023

Visiting Research Assistant

Berkeley, CA

- Implemented a reinforcement learning locomotion balancing controller for tailed quadruped robots using PyTorch and IsaacGym.

RoboDesign Lab: UIUC

Jan 2022 – May 2022

Undergraduate Research Assistant

Urbana, IL

- Estimated humanoid robot foot force signals using Gaussian processes with scikit-learn.

PUBLICATIONS

Cyrus Neary*, [Nathan Tsao](#)*, and Ufuk Topcu. Neural Port-Hamiltonian Differential Algebraic Equations for Compositional Learning of Electrical Networks. *Accepted to CDC 2025*.

Geffen Cooper*, [Nathan Tsao](#)*, Filippos Fotiadis, Ufuk Topcu, Radu Marculescu. Learning from Sparse and Asynchronous Data Streams for Batteryless Sensors. *In submission to SenSys 2026*.

TEACHING EXPERIENCE

ASE 370C Feedback Control Systems: UT Austin

Jan 2025 – May 2025

Graduate Teaching Assistant

Austin, TX

ME 314D Dynamics: UT Austin

Sep 2023 – Dec 2023

Graduate Teaching Assistant

Austin, TX